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Method for Post-Curing a Profile of Fibre-Reinforced Plastic Material

Abstract

The method involves supplying a profile which is wound round a winding structure with stresses being present in the profile due to the winding. The profile comprises several fibers extending along one another, which are embedded in a partially cured thermosetting matrix material. A heat treatment is carried out on the profile with a heat treatment device while the profile is wound. The matrix material is post-cured, during heat treatment. The glass-transition temperature of the matrix material is increased by the heat treatment, and the temperature to which the profile is exposed during the heat treatment remains below the glass-transition temperature. The stresses remain constant and the shape is retained both in cross-section as well as radius of curvature of the profile in the stress-free state, despite the heat treatment and despite the winding of the profile prior to the heat treatment.

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Background/Summary

[0001] This patent is a continuation of, and claims the benefit and priority of, co-pending U.S. patent application Ser. No. 18/733,747, filed Jun. 4, 2024, which is hereby incorporated by reference, which is a continuation of and claims the benefit and priority of U.S. patent application Ser. No. 18/123,271, filed Mar. 18, 2023, which is hereby incorporated by reference, which is a continuation of, and claims the benefit and priority of U.S. patent application Ser. No. 17/053,740, filed Nov. 6, 2020, which is hereby incorporated by reference, which is a 371 application of, and claims the benefit and priority of, international application PCT/EP2019/061837, filed May 8, 2019, which is hereby incorporated by reference, which claims foreign priority from NL application 2020895, filed May 18, 2018, which is hereby incorporated by reference.

[0002] The present invention relates to a method for post-curing a profile of fibre-reinforced plastic material.

[0003] It is known for profiles with a length of, for example, 1 or 2 metres, wherein said profiles are manufactured by the pultrusion process and their thermosetting matrix material is already partially cured in a heated die of the pultrusion machine during the pultrusion process, to be post-cured in an oven. By post-curing, the profile can be made more suitable for later applications at elevated temperatures, for example above 150° C. The applications in question may for example take place in a subsequent production process wherein the pultruded profile is processed as a semi-finished product at an elevated temperature. The known heat treatment has the result that the glass-transition temperature of the matrix material is increased.

[0004] The present invention aims to provide an improved method with which it is possible to produce profiles more efficiently, in particular (but not exclusively) to produce profiles in which the matrix material has an increased glass-transition temperature, for example a glass-transition temperature of at least 100° C. In addition, the present invention discerns that there is a market for pultruded profiles that have a relatively large length of, for example, tens or hundreds of metres. There may, for example, be an important advantage in that profiles of such lengths may be processed with a minimum amount of waste or that, broadly speaking, there can be continuous supply of the profile material to a production machine.

[0005] Within the context of the aforementioned aims, the method according to the present invention comprises the steps of: (a) supplying the profile, wherein the profile is wound around a winding body and wherein, as a result of the winding, stresses are present in the profile and said profile comprises a number of fibres extending along one another, which are embedded in a partially cured thermosetting matrix material, (b) by means of a heat treatment device, carrying out a heat treatment on the profile while the profile is wound around the winding body, and during said heat treatment the matrix material is post-cured, wherein the glass-transition temperature of the matrix material is increased as a result of the heat treatment and wherein the temperature to which

the profile is exposed during the heat treatment remains below the glass-transition temperature during the heat treatment, wherein the stresses remain constant and consequently the shape both in cross-section as well as radius of curvature of the profile is retained, at least in the stress-free state, despite the heat treatment and despite winding the profile round the winding body prior to the heat treatment.

[0006] The present invention is based moreover on the discernment that despite limited, incomplete, curing of the matrix material of the profile, it is possible to wind the profile around a winding body and then carry out a heat treatment on the wound profile, wherein further curing of the matrix material occurs, but without stresses that developed in the profile as a result of winding the profile round the winding body leading to permanent deformation of the profile after the heat treatment. This is achieved because the temperature to which the profile is exposed during the heat treatment remains below the glass-transition temperature, and said glass-transition temperature increases as a result of the heat treatment. The above reference to stress-free state is aimed at the situation in which the profile is unwound from the winding body after the heat treatment.

[0007] The present invention thus makes it possible to produce fibre-reinforced plastic profiles efficiently with relatively large lengths, for example hundreds of metres, and optionally with a relatively high glass-transition temperature.

[0008] The present invention is suitable in particular, but not exclusively, for application in the production of profiles which, in the stress-free state, extend linearly in the longitudinal direction thereof.

[0009] It may be advantageous if, for step a, the method according to the present invention comprises the step of winding the profile on the winding body. Winding the profile around the winding body may also be carried out at another location, for example in another factory, in another town or in another country, and by another party than at the location where the heat treatment takes place and then by the party carrying out the heat treatment.

[0010] The present invention is suitable in particular for application with pultruded profiles. Consequently, in a possible embodiment of the method according to the present invention, the profile is manufactured by a pultrusion process and step a comprises the steps of: (a-1) pulling fibres from a fibre supply unit by means of a pulling device, (a-2) applying the matrix material around the fibres by means of an applicator, (a-3) after step a-2, pulling the fibres pulled from the fibre supply unit through a heated die for partial curing of the matrix material in the die and shaping of the profile, (a-4) after step a-3, winding the profile on the winding body.

[0011] In order to make it possible for the location where the profile is wound around a winding body and the location where the heat treatment takes place to be different from one another, wherein for example transport of the profile when wound round the winding body from one location to the other location is necessary, the method according to the present invention may comprise the step of applying a separation in the partially cured profile at a position between the winding body and the heated die.

[0012] Alternatively, the present invention also offers the possibility for the profile to be continuous from the fibre supply unit to the heat treatment device.

[0013] A suitable manner of embedding the fibres in the matrix material may be obtained if the applicator according to step a-2 of the present invention comprises a bath for the matrix material in liquid form and the fibres pulled from the fibre supply unit are pulled through the liquid matrix material in the bath or if the applicator according to step a-2 comprises a bath for the matrix material in powder form and the fibres pulled from the fibre supply unit are pulled through the matrix material brought into a fluidized state.

[0014] An effective and efficient heat treatment of the profile may be obtained if the winding body with the profile wound around it is put in an oven during the heat treatment.

[0015] It was found that the heat treatment may be particularly effective if the heat treatment lasts at least 4 hours, more preferably at least 8 hours.

[0016] To promote the increase in the glass-transition temperature, it may be preferable for the heat treatment device to be controlled during the heat treatment in such a way that the temperature to which the profile is exposed during the heat treatment increases, in particular if the highest temperature to which the profile is exposed during the heat treatment is reached not earlier than 1.5 hours after the start of the heat treatment.

[0017] For increasing the glass-transition temperature it may further be advantageous if the highest temperature to which the profile is exposed during the heat treatment is at least 100° C., preferably at least 150° C., and more preferably at least 200° C.

[0018] The optimum settings of the aforementioned parameters will also depend on the dimensions of the profile and the nature of the matrix material used.

[0019] The profile may have favourable mechanical properties if the fibres are of carbon, glass, basalt, aramid or an oriented polymer. The fibres may for example be of quartz glass or E-glass. E-glass is alumina-lime-silicate glass that contains less than 1% w/w of alkali oxides. Especially favourable profile properties are obtained if the fibres are of carbon and/or a combination of quartz glass and aramid.

[0020] The profile may be very suitable for applications at elevated temperature, for example at 200° C. if the matrix material is a bismaleimide resin. Other suitable profiles may be obtained if the matrix material is an epoxy, polyester, vinyl ester, phenolic, polyurethane or silicone resin.

Furthermore, it has been discerned that profiles suitable for medical use may be obtained according to the present invention if the matrix material is an epoxy resin with a glass-transition temperature of at least 150° C.

[0021] In order to limit or prevent the development of internal mechanical stresses in the profile on account of thermal expansion of the winding body, the latter is preferably made of a material with a coefficient of thermal expansion less than $2.5 \cdot 10^{-6} \text{ K}^{-1}$. A suitable material is for example INVAR, which is a nickel-iron alloy with 64% iron, 35-36% nickel and small percentages of other elements such as manganese, carbon, chromium, magnesium, silicon and cobalt.

[0022] To prevent permanent deformation of the profile it may be advantageous for the windings of the profile round the winding body to be next to each other, in particular if the windings of the profile round the winding body do not touch each other. Mutual adhesion and/or damage of windings may thus also be avoided.

[0023] It may be advantageous, especially if the profiles are further processed at a location other than where they are produced, if the method according to the present invention comprises the further steps of: (c) unwinding the profile from the winding body after step b, and (d) winding the profile on another winding body after step c.

[0024] For transporting the profile it may then be possible to use a lower-grade winding body ("other winding body") than the winding body that is used during steps a, b and c.

[0025] It is then preferable if the other winding body according to step d has a smaller diameter than the winding body according to step c. The diameter of the winding body that is used during steps a, b and c may be adapted to the maximum bending that the profile can undergo during winding thereof around the winding body without breaking and optionally, for example, to the space that is available in an oven. The diameter of the other winding body may then be selected as smaller because the heat treatment has had the effect that the profile can withstand higher bending stresses. The property that the other winding body has a smaller diameter leads to the advantage that the other winding body with the profile wound thereon requires less space for example during storage and transport.

[0026] The present invention further relates to a profile manufactured by means of the method according to the present invention, wherein the fibres are of carbon and the matrix material is a bismaleimide resin. It has been discerned that such a profile has especially favourable properties for application as reinforcement of the thickness of a laminated material. Preferably the profile has a diameter of at most 0.5 millimetre.

[0027] The present invention further relates to a profile manufactured by means of the method according to the invention, wherein the fibres are of quartz glass and/or aramid and the matrix material is an epoxy resin with a glass-transition temperature of at least 150° C. It has been discerned that such a profile has especially favourable properties for medical applications, for example application as reinforcing material in a thermoplastic material, such as for guide-wires in minimally invasive surgery. Preferably the profile has a diameter of at most 0.5 millimetre.

[0028] The present invention is explained in more detail hereafter, referring to the figure, based on the description of a production line with which the methods according to the invention can be applied, and of the method itself.

Description

[0029] Accordingly, preferred embodiments of the present invention will now be described, by way of example only, with reference to the accompanying drawing, in which:

[0030] FIG. 1 is a schematic illustration of a pultrusion line according to the present invention.

[0031] The figure, FIG. 1, shows a pultrusion line 1 such as may be employed when carrying out a method according to the invention. Pultrusion line 1 comprises a number of winding bodies made in the form of bobbins 2 with rovings 3 wound thereon. Rovings are bundles of fibres, said fibres for example being of carbon or glass.

[0032] The pultrusion line 1 comprises a pulling device 4 with which the rovings 3 are pulled from the bobbins 2. In this example the pulling device comprises two endless belts 5. Each of the belts 5 is looped round two rotating members 6. A number of pressure rollers 7 are provided between the two rotating members. The rotating members 6 and the pressure rollers 7 belonging to the two belts 5 are directly opposite each other and incline, and/or are urged, towards each other under spring tension.

[0033] Downstream relative to the bobbins 2, the pultrusion line 1 comprises an alignment body 8a. Holes in the alignment body 8a position the rovings 3 at desired positions relative to, and at a distance from, each other.

[0034] Downstream relative to alignment body 8a, a bath 9 is provided. The bath 9 has four guide rolls 11, past which the rovings are led through matrix material in the form of liquid resin material in bath 9. The liquid resin is thus applied round each of the rovings 3.

[0035] Downstream relative to the bath 9, pultrusion line 1 is provided with a second alignment body 8b. Holes in the second alignment body 8b ensure that the rovings 3, on being enveloped in resin material, are maintained at the desired mutual positions after the bath 9. The pattern of holes in the second alignment body 8b does not have to be exactly identical to the pattern of holes in the first alignment body 8a.

[0036] The die orifice of the heated die 10 provided downstream relative to the second alignment body 8b has the shape of the profile 14 to be produced. The heating of the die 10 ensures that the resin is partially cured in the die 10 so that the profile 14, as it emerges downstream of the die, can be gripped by turns, e.g. two turns, of the belts 5 opposite to each other so that the pulling device 4 can pull the profile 14 forwards, thereby also, as stated earlier, pulling the rovings 3 from the bobbins 2.

[0037] Downstream of the pulling device 4 and aligned therewith, a winding device 12 is provided, with a winding body in the form of a spool 13. The spool 13 is made of a nickel-iron alloy that is also well known as INVAR and has a coefficient of thermal expansion of $1.2 \cdot 10^{-6} \text{ K}^{-1}$. In use, the profile 14 is wound by the winding device 12 on the spool 13, for example in lengths of tens to hundreds of metres. For this purpose, a separation is applied in profile 14 between spool 13 and pulling device 4, for example by means of shears or a saw. The diameter of the spool 13, and the heating of the profile 14 in the die 10, are selected and/or matched to one another so that the

profile **14** can withstand both the bending of the profile **14** that is necessary for winding, and the heat treatment that comes next.

[0038] In the next step, the spool **13** with the profile **14** wound round it is put in an oven for further curing to obtain the desired mechanical and thermal properties. The glass-transition temperature of the matrix material is increased as a result of this heat treatment.

[0039] An example of a possible embodiment of the method according to the present invention is described hereafter:

[0040] A number of rovings **3** with carbon fibres are pulled by the pulling device **4** with a speed of, for example, between 0.8 metre per minute and 1.4 metres per minute, through the heated die **10**. In bath **9** there is a resin, for example a high-grade resin based on bismaleimide. The rovings **3** are led through bath **9** and are formed in die **10** to a round shape (cross-section), wherein the rovings **3** are located at least essentially at a regular distance from each other and they are surrounded by the resin. In the die, the resin is partially cured at a temperature of approx. 180° C., and a pultruded round profile is formed. The profile is wound with the winding device **12** on the spool **13**, wherein the windings of the profile are located next to each other and are completely free from one another. After a desired length of the profile is wound on the spool, the profile **14** is sawn or cut through, between the pulling device **4** and the winding body **13**. Winding the profile **14** around the spool **13** causes mechanical stresses in the profile **14**. These mechanical stresses ensure that, in the hypothetical case where, directly after winding the profile **14**, the profile were to be unwound from the spool **13** again, it would in the stress-free state assume the shape that the profile would have in a stress-free state preceding winding round the spool **13**. A stress-free state could be obtained by separating a limited length of the profile **14**, for example a length of, for example, 1 or 2 metres, and laying it on a flat substrate. In this stress-free state the profile can extend linearly in the longitudinal direction thereof.

[0041] In the next step of the production process according to the present invention, spool **13** with the profile wound round it is put in an oven and heated therein. At the start of the heat treatment, the glass-transition temperature of the matrix material, i.e. of the resin partially cured in die **10**, may typically be for example 120° C. A possible heating profile is, for example, stepwise heating, wherein for example in a period of 4 hours the oven is heated from room temperature, or in any case from a temperature below 120° C., to a temperature of 190° C. As the temperature in the oven rises, the glass-transition temperature of the profile, more specifically of the matrix material thereof, also increases, for example to 220° C. During the temperature rise in the oven, the temperature in the oven remains in every case lower than the glass-transition temperature of the matrix material, which also increases. After the aforementioned highest temperature is reached in the oven, this temperature is maintained for 8 hours. During this period the glass-transition temperature increases even further, for example to 250° C.

[0042] The temperature to which the profile **14** is exposed in the oven always remains below the glass-transition temperature of the resin during the heat treatment, and said glass-transition temperature increases as a result of the heat treatment. The thermal expansion of the spool **13** is such that this does not lead to extra (tensile) stresses in the profile **14**. Extra stresses of this kind could lead to permanent deformation of the profile **14**.

[0043] Next, cooling of the profile to room temperature takes place. This cooling may take place partially or completely in the oven, but also may be completely outside of it. After this heat treatment, a further curing of the resin has occurred. The present invention is not, however, limited to this heating profile-it also depends on the composition of the resin used.

[0044] If the profile **14** were to be unwound from the spool **13** after the heat treatment and were to be made stress-free, then the profile **14** would have the same cross-section and radius of curvature as the profile **14** had in the stress-free state directly prior to the heat treatment. The foregoing reference to radius of curvature does not suggest that there must always be a question of a definite curvature. Within the context of the present invention, the profile preferably expands linearly in its

longitudinal direction in the stress-free state. Because the temperature to which the profile **14** is exposed during the heat treatment remains below the (increasing) glass-transition temperature of the matrix material, the mechanical stresses that develop in the profile **14** on account of winding the profile **14** round the spool **13** will not undergo relaxation.

[0045] After the heat treatment, the profile **14** is unwound from spool **13** and wound onto another, lower-grade, spool (“other winding body”) with a smaller diameter, for further transport and processing.

[0046] The present invention is explained above, by way of example only, on the basis of an example with a profile that has bismaleimide resin as matrix material. The present invention may, however, also be applied advantageously for producing profiles with a lower-grade matrix material, for example based on an epoxy resin. Such profiles can then be pultruded at a relatively high pultrusion speed, for example a speed of between 1.75 metres per minute and 2.75 metres per minute, without requiring a longer heated die. Owing to the relatively short residence time in the heated die, the resultant cure will still be incomplete. Then, by winding the pultruded profile onto a winding body and then subjecting the profile, wound round the winding body, to a further heat treatment, for example in an oven, further curing of the matrix material may take place, wherein in addition the profile becomes available in a single long length of, for example, some hundreds of metres.

[0047] In a particular embodiment according to the present invention, it is also possible for the steps of applying the resin round the rovings, and then winding thereof round a spool, to be carried out at some other location, for example by some other company, than the location where the heat treatment is carried out.

[0048] The present invention has been described above as having beneficial application in a medical device or application incorporating a profile manufactured according to the present invention. This is just one example, and the present invention also has beneficial application in civil engineering, automotive, marine, aeronautical and space applications, where profiles manufactured according to the present invention provide advantages over prior art profiles when used in apparatus, devices, parts, structures and manufacturing, building or assembly processes in these fields.

[0049] Although preferred embodiments of the present invention have been described above, any one or more or all of the features described (and/or claimed in the appended claims) may be provided in isolation or in any other combination or combinations in any of the embodiments, or in further embodiments. As such, any one or more the features may be removed, substituted and/or added to any of the individual features or feature combinations described and/or claimed. For the avoidance of doubt, any one of the features described herein may be presented individually, or combined with any other one or more features, even to the extent of creating a new embodiment.

[0050] Accordingly, whilst preferred embodiments of the present invention have been described above and illustrated in the drawing, these are by way of example only and non-limiting. It will be appreciated by those skilled in the art that many alternatives are possible within the ambit and scope of the invention, as set out in the appended claims.

[0051] The following clauses describe further preferred aspects of the present invention: [0052] 1. Method for post-curing a profile of fibre-reinforced plastic material comprising the steps of (a) supplying the profile, wherein the profile is wound round a winding body and wherein, as a result of the winding in the profile, stresses are present in the profile and said profile comprises a number of fibres extending along one another, which are embedded in a partially cured thermosetting matrix material, (b) by means of a heat treatment device, carrying out a heat treatment on the profile while the profile is wound round the winding body, and during said heat treatment the matrix material is post-cured, wherein the glass-transition temperature of the matrix material is increased as a result of the heat treatment and wherein the temperature to which the profile is exposed during the heat treatment remains below the glass-transition temperature during the heat

treatment wherein the stresses remain constant and as a result the shape is retained both in cross-section as well as radius of curvature of the profile, at least in the stress-free state, despite the heat treatment and despite winding the profile round the winding body prior to the heat treatment.

[0053] 2. Method according to clause 1, wherein the profile expands linearly in the stress-free state in the longitudinal direction thereof. [0054] 3. Method according to clause 1, wherein the method comprises, for step a, the step of winding the profile on the winding body. [0055] 4. Method according to one of clauses 1 or 2, wherein the profile is manufactured by a pultrusion process and step a comprises the steps of: (a-1) pulling fibres from a fibre supply unit by means of a pulling device, (a-2) applying the matrix material round the fibres by means of an applicator, (a-3) after step a-2, pulling the fibres pulled from the fibre supply unit through a heated die for partial curing of the matrix material in the die and shaping of the profile, (a-4) after step a-3, winding the profile on the winding body. [0056] 5. Method according to clause 4, wherein the method comprises the step of applying a separation in the partially cured profile at a position between the winding body and the heated die. [0057] 6. Method according to clause 4, wherein the profile is continuous from the fibre supply unit to the heat treatment device. [0058] 7. Method according to clause 4, 5 or 6, wherein the applicator comprises a bath for the matrix material in liquid form and the fibres pulled from the fibre supply unit are pulled through the liquid matrix material in the bath. [0059] 8. Method according to clause 4, 5 or 6, wherein the applicator comprises a bath for the matrix material in powder form and the fibres pulled from the fibre supply unit are pulled through the matrix material brought into a fluidized state. [0060] 9. Method according to one of the preceding clauses, wherein the winding body with the profile wound round it is put in an oven during the heat treatment. [0061] 10. Method according to one of the preceding clauses, wherein the heat treatment lasts at least 4 hours, more preferably at least 8 hours. [0062] 11. Method according to one of the preceding clauses, wherein the heat treatment device is controlled in such a way during the heat treatment that the temperature to which the profile is exposed during the heat treatment increases. [0063] 12. Method according to one of the preceding clauses, wherein the highest temperature to which the profile is exposed during the heat treatment is at least 100° C., preferably at least 150° C. and more preferably at least 200° C. [0064] 13. Method according to one of the preceding clauses, characterized in that the highest temperature to which the profile is exposed during the heat treatment is reached not earlier than 1.5 h after the start of the heat treatment. [0065] 14. Method according to one of the preceding clauses, wherein the fibres are of carbon, glass, basalt, aramid or an oriented polymer. [0066] 15. Method according to one of the preceding clauses, wherein the matrix material is an epoxy, polyester, vinyl ester, phenolic, polyurethane or silicone resin and preferably a bismaleimide resin. [0067] 16. Method according to one of the preceding clauses, wherein the spool is made of a material with a coefficient of thermal expansion less than $2.5 \cdot 10^{-6} \text{ K}^{-1}$. [0068] 17. Method according to one of the preceding clauses, wherein the windings of the profile round the winding body lie next to each other. [0069] 18. Method according to clause 17, wherein the windings of the profile round the winding body do not touch each other. [0070] 19. Method according to one of the preceding clauses comprising the further steps of: (c) after step b, unwinding the profile from the winding body and (d) after step c, winding the profile onto another winding body. [0071] 20. Method according to clause 19, wherein the other winding body according to step d has a smaller diameter than the winding body according to step c. [0072] 21. Profile manufactured by the method according to one of the preceding clauses, wherein the fibres are of carbon and the matrix material is a bismaleimide resin. [0073] 22. Profile manufactured by the method according to one of the preceding clauses, wherein the fibres are of quartz glass and/or aramid and the matrix material is an epoxy resin with a glass-transition temperature of at least 150° C. [0074] 23. Profile according to clause 21 or 22, wherein the diameter of the profile is at most 0.5 millimetre. [0075] 24. Use of the profile according to clause 21, 22 or 23 for application as reinforcing material in a thermoplastic material.

Claims

1. A method for post-curing a profile of fibre-reinforced plastic material comprising the steps of: a. supplying the profile, wherein the profile is wound round a winding body, and wherein, as a result of the winding in the profile, stresses are present in the profile and said profile comprises a number of fibres extending along one another, which are embedded in a partially cured thermosetting matrix material; and b. by means of a heat treatment device, carrying out a heat treatment on the profile while the profile is wound round the winding body, and during said heat treatment the matrix material is post-cured, wherein the glass-transition temperature of the matrix material is increased as a result of the heat treatment and wherein a temperature to which the profile is exposed during the heat treatment remains below the glass-transition temperature during the heat treatment wherein the stresses remain constant and as a result the shape is retained both in cross-section as well as radius of curvature of the profile, at least in a stress-free state, despite the heat treatment and despite winding the profile round the winding body prior to the heat treatment.
2. A method according to claim 1, wherein the profile expands linearly in the stress-free state in a longitudinal direction thereof, and/or the profile extends linearly in the stress-free state in a longitudinal direction thereof.
3. A method according to claim 1, wherein the method comprises, for step a, the step of winding the profile on the winding body.
4. A method according to any one of claim 1 or 2, wherein the profile is manufactured by a pultrusion process and step a comprises the steps of: a-1. pulling fibres from a fibre supply unit by means of a pulling device; a-2. applying the matrix material round the fibres by means of an applicator; a-3. after step a-2, pulling the fibres pulled from the fibre supply unit through a heated die for partial curing of the matrix material in the die and shaping of the profile; a-4. after step a-3, winding the profile on the winding body.
5. A method according to claim 4, wherein the method comprises the step of applying a separation in the partially cured profile at a position between the winding body and the heated die.
6. A method according to claim 4, wherein the profile is continuous from the fibre supply unit to the heat treatment device.
7. A method according to claim 4, wherein the applicator comprises a bath for the matrix material in liquid form and the fibres pulled from the fibre supply unit are pulled through the liquid matrix material in the bath.
8. A method according to claim 4, wherein the applicator comprises a bath for the matrix material in powder form and the fibres pulled from the fibre supply unit are pulled through the matrix material brought into a fluidized state.
9. A method according to claim 4, wherein the winding body with the profile wound round it is put in an oven during the heat treatment.
10. A method according to claim 1, wherein the heat treatment lasts at least 4 hours.
11. A method according to claim 1, wherein the heat treatment device is controlled in such a way during the heat treatment that the temperature to which the profile is exposed during the heat treatment increases.
12. A method according to claim 1, wherein a highest temperature to which the profile is exposed during the heat treatment is at least 100° C.
13. A method according to claim 1, characterised in that a highest temperature to which the profile is exposed during the heat treatment is reached not earlier than 1.5 hours after the start of the heat treatment.
14. A method according to claim 1, wherein the fibres comprise any one or more of carbon, glass, basalt, aramid or an oriented polymer.
15. A method according to claim 1, wherein the matrix material comprises any one or more of an

epoxy, polyester, vinyl ester, phenolic, polyurethane or silicone resin, a bismaleimide resin.

16. A method according to claim 1, wherein the winding body is made of a material with a coefficient of thermal expansion less than $2.5 \times 10^{-6} \text{ K}^{-1}$.

17. A method according to claim 1, wherein the windings of the profile round the winding body lie next to each other.

18. A method according to claim 1, wherein the windings of the profile round the winding body do not touch each other.

19. A method according to claim 1, comprising the further steps of: c. after step b, unwinding the profile from the winding body and d. after step c, winding the profile onto an other winding body.

20. A method according to claim 19, wherein the other winding body according to step d has a smaller diameter than the winding body according to step c.

21. A method according to claim 1, wherein the fibres are of carbon and the matrix material is a bismaleimide resin.

22. A method according to claim 1, wherein the fibres are of quartz glass and/or aramid and the matrix material is an epoxy resin with a glass-transition temperature of at least 150°C .

23. A method according to claim 1, wherein the diameter of the profile is at most 0.5 millimetre.

24. A method for post-curing a profile of fibre-reinforced plastic material comprising the steps of: a. supplying the profile, which profile adopts a substantially linear configuration in the longitudinal direction thereof when in a natural, relaxed or stress-free state, in a non-linear configuration in the longitudinal direction thereof where, as a result of supplying the profile arranged out of its linear configuration in the longitudinal direction thereof, stresses are present in the profile not otherwise present when in the natural, relaxed or stress-free state, and wherein said profile comprises a number of fibres extending along one another, which are embedded in a partially cured thermosetting matrix material; and b. by means of a heat treatment device, carrying out a heat treatment on the profile while the profile is in the non-linear configuration in the longitudinal direction thereof, and during said heat treatment the matrix material is post-cured, wherein the glass-transition temperature of the matrix material is increased as a result of the heat treatment and wherein a temperature to which the profile is exposed during the heat treatment remains below the glass-transition temperature of the matrix during the heat treatment.

25. A method according to claim 24, wherein, despite the heat treatment and despite the profile being in a non-linear configuration in the longitudinal direction thereof prior to and during the heat treatment, the shape of the profile prior to heat treatment is retained when in the natural, relaxed or stress-free state after heat treatment.

26. A method according to claim 1, wherein the heat treatment lasts at least 8 hours.

27. A method according to claim 1, wherein a highest temperature to which the profile is exposed during the heat treatment is at least 150°C .

28. A method according to claim 1, wherein a highest temperature to which the profile is exposed during the heat treatment is at least 200°C .
